K M Ahbab Zaman Aqib

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EDUCATION

Sylhet Agricultural University, Sylhet-3100, Bangladesh **CGPA-3.38**
B.Sc. in Agriculture *June 2022 - Ongoing*

Dhaka Residential Model College, Dhaka-1207, Bangladesh **GPA-5.00**
Higher Secondary Certificate *March 2020 - June 2022*

Nabin Chandra Model High School, Kulaura-3230, Bangladesh **GPA-5.00**
Secondary School Certificate *January 2018 - March 2020*

TECHNICAL SKILLS

Programming & Data Analysis

Programming Languages: R Programming Language – Data Manipulation, Statistical analysis, ggplot2 visualizations, dplyr, tidyr

Statistical Analysis – ANOVA, Regression, Correlation, Principal Component Analysis (PCA), Descriptive Statistics

Microsoft Excel – Data entry, Pivot tables, Charts, Basic Statistical Functions

Bioinformatics Tools & Databases

NCBI (GenBank, BLAST) – sequence retrieval and homology searching

MEGA (Molecular Evolutionary Genetics Analysis) – Phylogenetic Tree Construction

Primer3 / SnapGene – Primer design and sequence visualization

UniProt / Ensembl Plants – Protein and genome database querying

KEGG / Gene Ontology (GO) – pathway and functional annotation analysis

Motif Discovery and Conserved Motif Analysis (MEME Suite)

Protein 3D Structure Prediction and Visualization (AlphaFold, SWISS-MODEL, PyMOL)

Molecular Docking Analysis (AutoDock Vina, PyRx, CB-Dock2)

Cis-Regulatory Element Analysis and Promoter Characterization (PlantCARE)

Laboratory & Field Skills

Physiological Assays – Chlorophyll Content, Relative Water Content (RWC), Electrolyte Leakage

Biochemical Assays – Proline Content, MDA (Lipid Peroxidation), Antioxidant Enzyme Activity (SOD, POD, CAT)

Salinity Stress Experiment Design and Controlled Greenhouse Management

Molecular Biology Techniques – Genomic DNA Extraction using CTAB Method from Plant Tissues

Germination Bio-Assay, seedling growth analysis, plant Phenotyping

PROFESSIONAL EXPERIENCE

Undergraduate Researcher (Self-Directed)

Department of Agronomy, Sylhet Agricultural University, Sylhet-3100, Bangladesh *2024 - Present*

- Conducting physiological and biochemical characterization of contrasting rice varieties under salinity stress conditions.
- Measuring key stress-tolerance indicators including proline accumulation, MDA levels, and antioxidant enzyme activities (SOD, POD, CAT).
- Collecting and analyzing experimental data using R programming; applying ANOVA and descriptive statistics for result interpretation.
- Reviewing current literature on abiotic stress tolerance mechanisms in rice, particularly in the context of climate-resilient crop development.
- Supervisor / Faculty Advisor: Professor Dr. Mozammel Hoque, Department of Agronomy, Sylhet Agricultural University.

WORKING EXPERIENCE

Biology Instructor
Phoenix Coaching Centre, Kulaura, Sylhet, Bangladesh *2024 - Present*

- Teaching Biology to SSC and HSC level students, covering cell biology, genetics, plant physiology, and ecology.
- Preparing lesson plans, study materials, and model test papers aligned with the national curriculum.
- Mentoring students on exam preparation strategies, resulting in improved academic outcomes.
- Communicating complex biological concepts in simplified, engaging formats appropriate for the age group.

PROJECTS

- **Physiological and Biochemical Characterization of Contrasting Rice Varieties Subjected to Salinity Stress**, *Department of Agronomy, Sylhet Agricultural University, (Ongoing research project)*. GitHub
  - Crop Stress Tolerance and Improvement (Ongoing)
  - Investigating the differential salinity stress responses of salt-tolerant and salt-sensitive rice genotypes at seedling stage.
  - Measuring physiological parameters: shoot/root length, fresh/dry weight, relative water content (RWC), SPAD chlorophyll index.
  - Assessing biochemical markers: proline content, lipid peroxidation (MDA), and antioxidant enzyme activity.
  - Performing statistical analysis of experimental data using R; comparing variety-level differences through ANOVA and post-hoc tests.
  - Findings expected to contribute to identifying stress-tolerance biomarkers for future breeding programs.

CONFERENCE PRESENTATIONS

- **Presenting Author – Poster Presentation**, *5th International Conference of the Bangladesh Society of Plant Science and Technology (BSPST), Sylhet Agricultural University, Sylhet, Bangladesh* — 29–30 October 2026
- **Poster Title:** Expression Dynamics of the Cytochrome B5 Gene Family in *Arabidopsis thaliana*: Tissue Specificity, Phytohormonal Responsiveness, and Abiotic Stress Responses.
- Collaborative research poster presented by me as the Presenting Author.

PUBLICATIONS

- *No peer-reviewed publications at present. Research manuscript in preparation based on ongoing project findings. Manuscript in Preparation*: "Physiological and Biochemical Responses of Contrasting Rice Varieties under NaCl-Induced Salinity Stress" Link
- *Research manuscript in preparation based on ongoing project findings. Manuscript in Preparation*: "Genome-Wide Identification and Characterization of the Rad5/16-like Group Gene Family in Arabidopsis thaliana with Their Developmental, Hormonal, and Stress-Responsive Expression" {Link

RESEARCH INTEREST

- Abiotic Stress Tolerance in Crops – particularly salinity, drought, and heat stress mechanisms in rice and other cereal crops
- Plant Molecular Biology and Functional Genomics – gene expression, QTL mapping, marker-assisted selection (MAS)
- Plant Bioinformatics & Computational Biology – genome-wide association studies (GWAS), omics data integration, comparative genomics
- Climate-Resilient Agriculture & Food Security – developing stress-tolerant crop varieties for tropical and subtropical environments
- Agronomy & Crop Physiology – understanding crop growth, yield formation, and nutrient use efficiency
- Sustainable Agriculture & Soil Health – precision agriculture, soil microbiome interactions, and eco-friendly crop management

CERTIFICATION

- From inbox to impact: Improve your email workflows with AI Github
- Make your meetings more productive with AI Github
- Uncover new data insights with AI Github
- Unlock productivity and unleash creativity with AI powered chat Github

EXTRACURRICULAR ACTIVITIES

- **President,** Kulaura Students Association, Sylhet — (2026 - Present)
- **Executive Committee Member,** Public University Students Association of Kulaura — (2024 - Present)
- **Active Scout Member,** Bangladesh Scouts – Participated in the 10th Bangladesh National Scout Jamboree and the 3rd SAANSO Scout Jamboree (South Asia and Arabian Nations Scout Organization); also participated in Zilla Scout Camps and various district-level scouting activities.
- **Social Organization Volunteer,** Engaged with multiple community-based social organizations focused on youth development, education awareness, and community service in Kulaura, Moulvibazar.

REFERENCE

- **Dr. Md. Mozammel Haque**
  Professor
  Department of Agronomy, Sylhet Agricultural University, Sylhet-3100, Bangladesh.
  Email: hoquem.aha@sau.ac.bd
- **Dr. Sayeda Sultana**
  Professor
  Department of Genetics & Plant Breeding, Sylhet Agricultural University, Sylhet-3100, Bangladesh.
  Email: sultana.gpb@sau.ac.bd